

Image-to-Image Translation with Pix2Pix on the Facades Dataset

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Abstract

In this project, we use the Facades dataset to perform image-to-image translation. Our goal is to take simple color-coded maps of building layouts and turn them into highly realistic photos. To achieve this, we set up a Pix2Pix model, a conditional Generative Adversarial Network (CGAN) that pairs together a U-Net generator and a PatchGAN discriminator guided by an L1 reconstruction loss. A baseline model is trained from scratch (Task 1) and then improved with data augmentation (Task 2), which increases the diversity of the small training set and mitigates overfitting. Models are evaluated with PSNR, SSIM, LPIPS and FID on a held-out test split, and a stand-alone inference demo is provided (Task 3).

1. INTRODUCTION

Image-to-image translation learns a mapping from an input image domain to an output image domain. In our case, the input is a discrete label map (a colour map of architectural elements such as windows, doors, walls and balconies) and the output must be a fully realized, photorealistic building facade.

We utilize the Pix2Pix architecture, where both the generator and discriminator are conditioned on the input label map. The generator takes a label map and produces a fake photo. It uses a U-Net architecture, an encoder-decoder with skip connections that lets low-level spatial details flow directly from the encoder to the decoder. The discriminator receives the label map together with a photo (real or generated) and must tell them apart. It is a PatchGAN: instead of one global decision it classifies many overlapping image patches, which encourages sharp, locally realistic textures.

The two networks are trained adversarially while an additional L1 loss keeps the generated photo structurally close to the ground truth. We build a faithful Pix2Pix baseline (Task 1) following the original paper: native 256×256 resolution, an 8-level U-Net, instance normalization, a learning-rate decay schedule and dropout used as a noise source at both train and test time. We then add data augmentation as the Task 2 improvement, and finally wrap the best generator into an inference program (Task 3).

2. TECHNICAL APPROACH

2.1. Dataset Configuration

We train and evaluate our models using the Facades dataset, which consists of 500 paired images of size 512×256 . Each file places the real photo on the left half and its label map on the right half. To properly evaluate our model's ability to generalize, we divided these 500 images into three distinct subsets: train (70%), validation (15%) and test (15%).

Our custom dataset class splits these pairs apart, resizes them to 256×256 , and scales the pixel values to a $[-1, 1]$ range to match the tanh activation function. Label maps are resized using nearest-neighbour interpolation to keep discrete colours, while real photos are resized using bicubic interpolation to maintain smooth transitions.

2.2. Baseline Architecture (Task 1)

For our baseline, we implemented the architecture as described in the original literature. The generator compresses the 256×256 input down to a 1×1 bottleneck before expanding it back. Skip connections join each encoder block to the matching decoder block. Dropout layers act as our primary source of stochastic noise. The discriminator evaluates the concatenation of the label map and the photo (creating a 6-channel input) to judge realism. We train this baseline without any data augmentation to establish a clear starting point for our metrics. Both networks use Adam optimization with a learning rate of $2e-4$ for 150 epochs.

2.3. Data Augmentation (Task 2)

A recurring concern with GANs is that the generator may overfit the training set and reproduce memorized copies of the training images rather than generalizing. Our Facades training split has only ~ 350 images, so overfitting is a real risk.

Since collecting new data is not possible, we increase the diversity of the training set synthetically through data augmentation. We apply the random jitter of the original Pix2Pix paper:

- **Up-scale + random crop:** images are resized to 286×286 and then cropped back to 256×256 at a random position.
- **Random horizontal flip:** the pair is mirrored with probability 0.5.

The label map and the photo always receive the exact same transformation to preserve spatial alignment.

2.4. Demo and Repository (Task 3)

The best performing generator is exported to float16 format to maintain a lightweight file size. A standalone inference script was developed, allowing users to process new label maps directly from the command line. The complete code and deliverables are available in our public GitHub repository.

3. RESULTS

We evaluate four complementary metrics on the held-out test split: PSNR and SSIM (higher is better) for structural fidelity, and LPIPS and FID (lower is better) for perceptual and statistical realism.

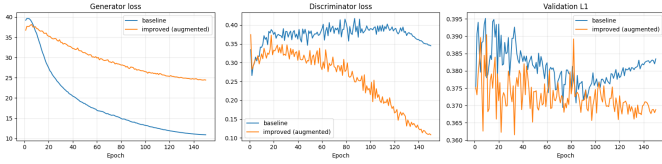


Figure 1: Comparison of Generator, Discriminator, and Validation L1 loss curves between the baseline and improved augmented model.

3.1. Training Curves Comparison Analysis

By plotting the training history of our baseline model against the improved model trained with data augmentation, we clearly see how synthetic dataset expansion changed the entire learning dynamic:

- **Generator Loss:** Unlike the baseline curve which drops rapidly, the augmented model’s generator loss decreases at a much slower and more gradual rate. Data augmentation continuously forces the generator to face newly cropped and mirrored variations of the images, slowing down the loss drop but ensuring it actually learns generalized features instead of memorizing shortcuts.
- **Validation L1 Loss:** While the baseline curve shows a classic overfitting “U-shape” that climbs steadily after epoch 80, the augmented model’s validation loss remains consistently lower and continues a steady downward trend all the way to epoch 150. This confirms that our Task 2 modifications drastically improved its ability to generalize to unseen data.

3.2. Metrics Comparison

By introducing data augmentation, we achieved noticeable improvements across every single performance metric compared to our baseline results, as detailed in Table 1.

The most impressive result is the massive reduction in FID, which plummeted by -44.8821 points. Because FID measures the statistical distance between the real and generated image distributions, this major change confirms that data augmentation allowed the generator to capture real-world facade styles and produce much cleaner, high-quality textures.

Table 1: Test Metrics Comparison

Metric	Baseline	Improved	Change
PSNR	12.3209	13.0348	+0.7138
SSIM	0.2049	0.2543	+0.0493
L1	0.1901	0.1723	-0.0178
LPIPS	0.5543	0.5146	-0.0397
FID	180.4805	135.5984	-44.8821

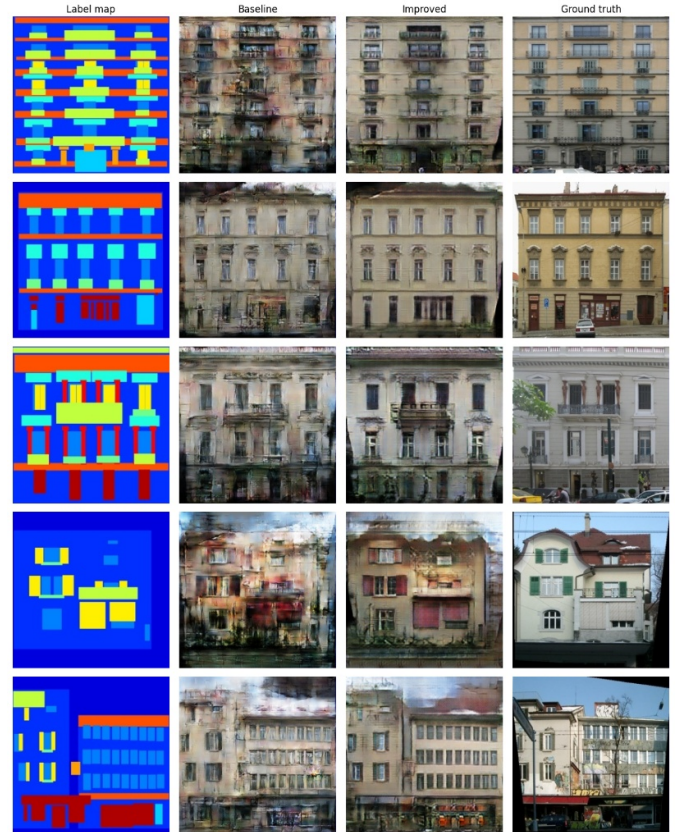


Figure 2: Side-by-side test outputs comparing Label Maps, Baseline Generated, Improved Generated, and Ground Truth photos.

3.3. Qualitative Comparison Analysis

To visually understand the impact of data augmentation, we compared side-by-side test outputs. The visual differences clearly show why the numerical metrics improved so much:

- **Sharper and Cleaner Structures:** The improved model outputs are noticeably crisper and more realistic than the baseline, which often looked like a blurry, smeared painting with wavy window edges.
- **Massive Reduction in Artifacts:** The baseline images suffer from random gray smudges, color bleeding, and weird artifacts. The improved model completely cleans up these noisy patches, creating uniform wall textures and smooth color fields.
- **Better Object Realism:** The baseline occasionally hallucinates chaotic mixes of colors. The improved model handles complex layouts beautifully, producing structured windows and solid wall colors that closely follow the real house’s appearance.

4. CONCLUSION

In this project, we successfully implemented a Pix2Pix conditional GAN to perform image-to-image translation on the Facades dataset, turning flat architectural blueprints into realistic building photographs.

Task 1 gave us a solid baseline using an 8-level U-Net generator and a PatchGAN discriminator trained at the native 256×256 resolution. While it learned the basic geometric layouts perfectly, it suffered from severe overfitting due to our small 350-image training split. Task 2 directly fixed this problem by adding data augmentation, including random scaling, cropping, and horizontal flipping. This forced the model to generalize, leading to massive improvements across the board—most notably dragging our FID score down by a huge 44.88 points. Finally, Task 3 allowed us to take our optimal generator, convert its weights to float16, and package it into a standalone inference script inside a ready-to-publish GitHub repository folder.

4.1. Limitations and Future Work

While our improved model achieved a massive performance jump, it still has a few clear limitations. The generator still struggles with micro-level texture details, meaning that fine elements like individual bricks, stone masonry, and intricate balcony railings still appear slightly smoothed out compared to the ground truth. Additionally, in complex label maps, the model can introduce minor blur around boundaries, proving that standard adversarial loss and L1 loss aren’t fully capable of forcing the model to learn high-frequency photographic details.

Future strategies could include incorporating a perceptual loss term using a pretrained VGG network to force the generator to match high-level visual features. Upgrading to a multi-scale architecture like Pix2PixHD would allow the network to handle fine textures at higher resolutions seamlessly.

References

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